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Finance and Investments club

Investment Banking Primer

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FOREWORD

Greetings!

The Finance and Investment Club of IIM Calcutta is excited to present the first edition of Investment Banking Primer.

The motive behind this primer is to explain important concepts asked in investment banking interviews. We advise looking at the Corporate Finance Primer too to understand some basic finance concepts.

Along with the primers, we would advise looking at the Finance Interview Question Bank and other materials shared by the club for preparation. Lastly, reach out to as many investment banking interns as you can to understand their interview and internship experiences.

We hope that this primer helps in bringing you one step closer to landing your dream internship. Feel free to reach out to us for any queries/suggestions.

We wish you all the very best!

Finance and Investments Club

IIM Calcutta

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FINANCIAL RATIOS

What are Financial Ratios?

Financial ratios are mathematical comparisons of financial statement accounts or categories. These relationships between the financial statement accounts help investors, creditors, and internal company management understand how well a business is performing and of areas needing improvement.

Ratios are just a raw computation of financial position and performance. In a sense, financial ratios don't take into consideration the size of a company or the industry.

Financial ratios are grouped into the following categories:

- Liquidity ratios
- Solvency ratios
- Efficiency ratios
- Profitability ratios
- Market value ratios

Liquidity Ratios

Liquidity of a company depends on its ability to convert assets into cash or to generate cash through operations to meet short term obligations.

Some important liquidity ratios are as follows –

1. **Current Ratio**: The current ratio measures a company's ability to pay off its short-term liabilities with current assets. Higher the current ratio, greater is the assurance that current liabilities will be met. A thumb rule of 2 is widely used as an optimal ratio.

Current ratio = Current assets / Current liabilities

2. **Quick Ratio/Acid test Ratio**: It measures a company's ability to pay off short-term liabilities with quick assets. It is a more stringent test of liquidity. It includes only those current assets which can be readily converted into cash. Inventories are excluded because they are least liquid and prepaid expenses are excluded because they will not get converted into cash.

Acid-test ratio = (Current assets – Inventories – Prepaid Expenses) / Current liabilities

3. **Cash Ratio**: The cash ratio measures a company's ability to pay off short-term liabilities with cash and cash equivalents. It is more stringent in comparison to the quick ratio. It emphasises that ultimately cash is required to settle current liabilities.

Cash ratio = (Cash and Cash equivalents + Marketable Securities)/ Current Liabilities

Note: A higher current ratio is more favourable than a lower current ratio because it shows the company can more easily make current debt payments. But high current ratio also shows underutilization of resources which could have been used to increase profitability of the company by investing them elsewhere.

Solvency Ratios

Solvency refers to the long-term viability of the company. Ratios related to capital structure and earnings coverage are used to measure solvency. It measures how leveraged the company is and how well is it placed with respect to its debt repayments capacity.

Some important solvency ratios are as follows -

1. **Debt to Total Assets Ratio**: It measures the relative amount of a company's assets that are provided from debt. Total liabilities include both current and non-current liabilities.

Debt ratio = Total liabilities / Total assets

2. **Debt to Equity Ratio**: It calculates the weight of total debt and financial liabilities against shareholders' equity.

Debt to equity ratio = Total liabilities / Shareholder's equity

3. **Interest Coverage Ratio**: It shows how easily a company can pay its interest expenses on the assumed debt. Higher interest coverage ratios imply the greater ability of the firm to pay off its interests. If Interest coverage is less than 1, then EBIT is not sufficient to pay off interest, which implies finding other ways to arrange funds.

Interest coverage ratio = EBIT/ Interest expenses

4. **Debt Service Coverage Ratio**: It reveals how easily a company can pay its debt obligations. Total debt service includes Principal payment, Interest payment and Lease payment.

Debt service coverage ratio = Operating income / Total debt service

Turnover Ratios

The turnover ratio is also known as **activity ratio**. This type of ratio indicates the efficiency with which an enterprise's resources are utilized.

Some important turnover/activity ratios are as follows -

1. **Inventory Turnover Ratio**: It measures how many times a company's inventory is sold and replaced over a given period. It is a measure of how efficiently a company can control its merchandise, so it is important to have a high turnover ratio.

Inventory turnover ratio = Cost of goods sold / Average inventory

Days sales in inventory = 365 days / Inventory turnover ratio

2. **Accounts Receivables turnover ratio**: It measures a business' ability to efficiently collect its receivables. Higher ratios mean that companies are collecting their receivables more frequently throughout the year. The reason net credit sales are used is because only credit sales establish a receivable.

Receivables turnover ratio = Net credit sales / Average accounts receivable

Days Accounts Receivables = 365/ Receivables turnover Ratio

3. **Accounts Payables turnover ratio**: It measures a business' ability to efficiently pay its receivables. Lower ratios mean that companies have more time to pay their payables throughout the year.

Payables turnover ratio = Net credit Purchases / Average accounts payable

Days Accounts Payables = 365/ Payables turnover Ratio

4. **Asset Turnover Ratio**: It measures a company's ability to generate sales from its assets. The total asset turnover ratio calculates net sales as a percentage of assets to show how many sales are generated from each dollar of company assets.

Asset turnover ratio = Net sales / Average total assets

5. **Cash Conversion Cycle**: The cash conversion cycle is a cash flow calculation that attempts to measure the time it takes a company to convert its investment in inventory and other resource inputs into cash. In other words, the cash conversion cycle calculation measures how long cash is tied up in inventory before the inventory is sold and cash is collected from customers.

Cash Conversion cycle = Days sales in inventory + Days Accounts Receivables - Days Accounts Payables

Profitability Ratios

Profitability ratios measure a company's ability to generate income relative to revenue, balance sheet assets, operating costs, and equity.

Common profitability financial ratios include the following -

1. **Gross Margin Ratio**: It compares the gross profit of a company to its net sales to show how much profit a company makes after paying its cost of goods sold. It basically shows % amount of money from product sales left over after all of the direct costs associated with manufacturing the product has been paid.

Gross margin ratio = Gross profit / Net sales

2. **Operating Margin Ratio**: It compares the operating income of a company to its net sales to determine operating efficiency. Many times, operating income is classified as earnings before interest and taxes. If companies can make enough money from their operations to support the business, the company is usually considered more stable.

Operating margin ratio = Operating income / Net sales

3. **Return on Assets Ratio**: The return on assets ratio measures how effectively a company can earn a return on its investment in assets. In other words, ROA shows how efficiently a company can convert the money used to purchase assets into net income or profits.

Return on assets ratio = Net income / Total assets

4. **Return on Equity Ratio**: It measures how efficiently a company is using its equity to generate profit. In other words, the return on equity ratio shows how much profit each dollar of common stockholders' equity generates.

Return on equity ratio = Net income / Shareholder's equity

Market value ratios

Market value ratios are used to evaluate the share price of a company's stock. Common market value ratios include the following:

1. **Book Value per share Ratio**: It calculates the per-share value of a company based on equity available to common shareholders divided by the number of outstanding shares. This figure represents the minimum value of a company's equity and measures the book value of a firm on a per-share basis.

Book value per share ratio = (Shareholder's equity – Preferred equity) / Total common shares outstanding

2. **Dividend Yield Ratio**: It measures the amount of cash dividends distributed to common shareholders relative to the market value per share. The dividend yield is used by investors to show how their investment in stock is generating cash flows in the form of dividends.

Dividend yield ratio = Dividend per share / Share price

3. **Earnings per share Ratio**: It measures the amount of net income earned per share of stock outstanding. In other words, this is the amount of money each share of stock would receive if profits were distributed to the outstanding shares at the end of the year.

Earnings per share ratio = (Net income- Preferred Dividends) / Total shares outstanding

4. **Price-Earnings Ratio**: It calculates the market value of a stock relative to its earnings by comparing the market price per share by the earnings per share. In other words, the price earnings ratio shows what the market is willing to pay for a stock based on its current earnings.

Price-earnings ratio = Share price / Earnings per share

FINANCIAL STATEMENTS

What are the three main financial statements?

Income Statement - A company's income statement provides details on the revenue a company earns and the expenses involved in its operating activities. Overall, it provides more granular detail on the holistic operating activities of a company. Broadly, the income statement shows the direct, indirect, and capital expenses a company incurs.

Revenues – COGS – Expenses = Net Income

Balance Sheet - Also referred to as the statement of financial position, a company's balance sheet provides information on what the company is worth from a book value perspective. The balance sheet is broken into three categories and provides summations of the company's assets, liabilities, and shareholders' equity on a specific date

Assets = Liabilities + Shareholder's Equity

Statement of Cash Flows - The cash flow statement provides a view of a company's overall liquidity by showing cash transaction activities. It reports all cash inflows and outflows over the course of an accounting period with a summation of the total cash available.

Beginning Cash + CF from Operations + CF from Investing + CF from Financing = Ending Cash

How are the 3 statements connected?

These three statements are interrelated in several ways, as noted in the following ways:

- The net income figure in the income statement is added to the retained earnings line item in the balance sheet, which alters the amount of equity listed on the balance sheet.

- The net income figure also appears as a line item in the cash flows from operating activities section of the statement of cash flows.

Income Statement	2018	2019
in €m	Act	Act
Revenue	100	110
Costs of Goods Sold	(60)	(65)
Gross Profit	40	45
Operating Expenses	(23)	(20)
EBITDA	17	25
D&A	(2)	(2)
EBIT	15	23
Financial Result	(5)	(6)
Tax	(3)	(5)
Net Income	7	12

Balance Sheet	2018	2019
in €m	Act	Act
Cash	10	13
Accounts Receivables	15	19
Inventory	15	19
PP&E	20	19
Total Assets	**60**	**70**
Accounts Payable	10	12
Debt	20	16
Total Liabilities	**30**	**28**
Share Capital	23	23
Retained Earnings	7	19
Shareholders' Equity	**30**	**42**
Total Liabilities & Equity	**60**	**70**

Cash Flow Statement	2018	2019
in €m	Act	Act
Net Income		12
D&A		2
Δ Accounts Receivables		(4)
Δ Inventory		(4)
Δ Accounts Payable		2
Operating Cash Flow		**8**
Investments in PP&E		(1)
Investing Cash Flow		**(1)**
Debt Repayment		(4)
Financing Cash Flow		**(4)**
Total Change of Cash		**3**

- Changes in various line items in the balance sheet roll forward into the cash flow line items listed on the statement of cash flows. For example, an increase in the outstanding amount of a loan appears in both the liabilities section of the balance sheet (as an ongoing balance) and in the cash flows from financing activities section of the statement of cash flows (in the amount of the incremental change).

- The ending cash balance in the balance sheet also appears in the statement of cash flows.

- The purchase, sale, or other disposition of assets appears on both the balance sheet (as an asset reduction) and the income statement (as a gain or loss, if any).

In short, the financial statements are highly interrelated. Consequently, when reviewing the financial statements of an organization, one should examine all of the financial statements in order to obtain a complete picture of its financial situation.

Example 1. How would a $10 increase in depreciation expense affect the three financial statements?

- Start with the income statement
  - The $10 increase in depreciation is an expense, which therefore lowers your operating profit by $10 and you will pay less taxes.
  - Your taxes will decrease by ($10 x T) and therefore your net income will decrease by ($10 x (1- T))
  - Assuming a 40% tax rate, the drop in net income will be $6

- Then move on to the statement of cash flows
  - The reduction in net income reduces cash from operations by $6
  - However, since the depreciation is a non-cash item so it will increase cash from operations by $10 since you add back depreciation
  - Ending cash is therefore increased by $4

- Then to the balance sheet
  - Cash increases by $4
  - PP&E reduces by $10 because of the depreciation
  - Overall assets fall by $6
  - This needs to balance with the other side of the balance sheet. In order to balance the balance sheet, retained earnings will fall by $6 due to the drop in net income)

Example 2. Impact on three financial statements if a machine of $100 is sold for $80

- Start with the income statement
  - Loss on sale of machinery leads to decrease in other income by $20
  - Decrease in profits will lead to tax savings of 20*(T) = 4 (Assuming 20% tax rate)
  - Overall decrease in N.I = -20 + 4 = $16

- Then move on to the statement of cash flows

  Operating CF
  - The reduction in net income reduces cash from operations by $16
  - However, since the loss on sale of machinery is a non-operating cash item so it will increase cash from operations by $20
  - Thus, Operating CF increases by 20 – 16 = $4 (tax saving)

Investing CF

- Sale of machinery leads to increase in cashflow of $80
- Overall increase in cash flow = Investing CF + Operating CF = 80 + 4 = $84

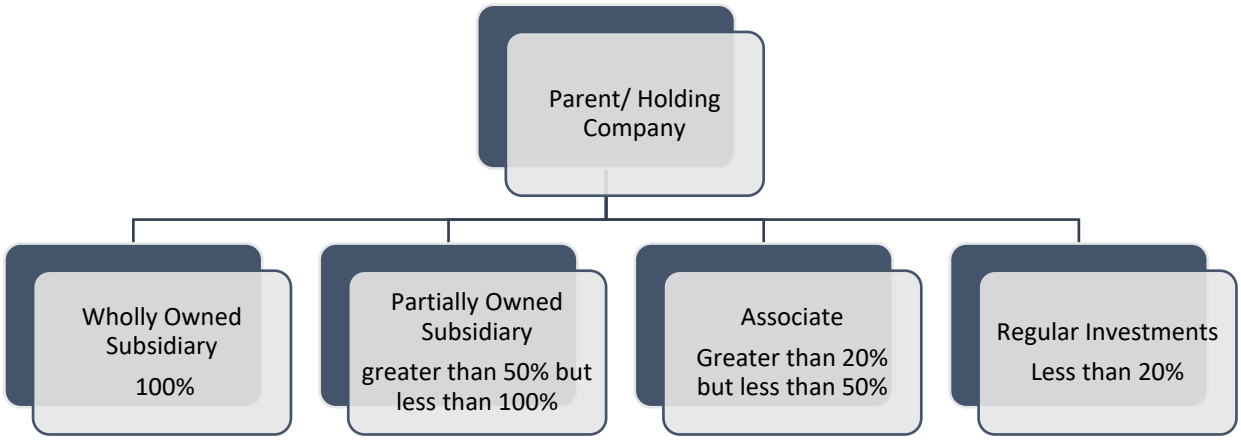
- Then to the balance sheet
  - Cash increases by $84
  - PP&E reduces by $100
  - Overall assets fall by $16
  - This needs to balance with the other side of the balance sheet. In order to balance the balance sheet, retained earnings will fall by $16 due to the drop in net income

CROSS HOLDINGS

Cross holding is a situation in which a publicly-traded corporation owns stock in another publicly-traded company. So, technically, listed corporations' own securities issued by other listed corporations. If the cross holding is not accounted for when determining the values of companies on the exchange, the result can be a double counting of the security's value, which would obscure the true value of the companies involved. It is also important to consider the role of cross holdings in situations like corporate takeovers and ousters of management.

In a simple example of a cross holding, Company X can hold stocks or bonds issued by Company Y. Company Y's value would be counted twice if the cross holding was not accounted for. It would be counted once when looking at the value of securities issued by the company, and again when considering the securities held by Company X. This would lead to having skewed information about the values of companies on the index.

Investment in another company can be in multiple forms-



Subsidiaries

In the corporate world, a subsidiary is a company that belongs to another company, which is usually referred to as the parent company or the holding company.

The parent holds a controlling interest in the subsidiary company, meaning it has or controls more than half of its stock. In cases where a subsidiary is 100% owned by another firm, the subsidiary is referred to as a wholly owned subsidiary.

How a Subsidiary Works

A parent company buys or establishes a subsidiary to provide the parent with specific synergies, such as increased tax benefits, diversified risk, or assets in the form of earnings, equipment, or property. Still, subsidiaries are separate and distinct legal entities from their parent companies, which reflects in the independence of their liabilities, taxation, and governance. If a parent company owns a subsidiary in a foreign land, the subsidiary must follow the laws of the country where it is incorporated and operates.

However, given their controlling interest parent companies often have considerable influence with their subsidiaries. They—along with other subsidiary shareholders, if any—vote to elect a subsidiary company's board of directors, and there may often be a board-member overlap between a subsidiary and its parent company.

To be designated a subsidiary, at least 50% of a firm's equity has to be controlled by another entity. If the stake is less than that, the firm is considered an associate or affiliate company. When it comes to financial reporting, an associate is treated differently than a subsidiary.

Consolidation entries for subsidiary

When the parent has legal control over the subsidiary, the parent will consolidate subsidiary financial statement. The consolidated financial statements are a combination of the subsidiary and parent financial statements.

Balance Sheet: The consolidated balance sheet will combine all assets and liabilities of the parent and subsidiary. We include all the balances even though the parent may not own 100% of the share. In the Equity part of the Balance sheet, a Non-Controlling Interest will appear which represents the share of minority shareholders.

Income Statement: We consolidate 100% of revenues and expenses into the consolidated income statement. The proportion of NCI net income will be subtracted to ensure that only the profit attributable to the parent will be show in the consolidated income statement.

Elimination Entries: These are adjusting entries which aim to eliminate duplicated balance in the consolidated financial statement. For example, subsidiary may have a balance with parent, so they both record Account Receivable and Account Payable. But when we consolidate, this balance must be eliminated; otherwise, we will overstate assets and liability. The same thing happens to revenue- the parent sells goods to the subsidiary, the parent will record revenue. Then subsidiary sells the same goods to third party, subsidiary will record revenue too. That is ok for the separate report, but in consolidate, we can’t record double revenue for the same goods.

Minority Interest/ Non-Controlling Interest

Minority interest is the value of a share, or the interest attributable to the shareholders holding less than 50% of the total number of shares. It is also known as non-controlling interest.

In the accounting world, it means ownership in a subsidiary company not owned by a holding company, which is also known as Parent Company. For a Company to be a holding company, it must always hold more than 50% of the shares in its subsidiary company.

For example, A & B are the two shareholders of Company Pine-Apple Inc., holding 80% and 20%, respectively. In the Balance-sheet of Pine –Apple Inc. Shareholder B will be considered as a minority shareholder since it owns less than 50% of total shares. Whereas, shareholder A is the majority shareholder of Pine-Apple Inc.

Minority interest adjustments occur only when the parent does not own 100% of the subsidiary.

Shareholders' Equity		
Common stock, $1 par value (2,000,000,000 shares authorized, 1,465,706,360 shares issued)	1,466	1,466
Additional paid-in capital	1,438	1,236
Retained earnings	18,861	18,832
Accumulated other comprehensive income (loss)	(3,950)	(3,507)
Unearned compensation	(12)	(20)
Treasury stock, at cost	(18,102)	(16,862)
Total Colgate-Palmolive Company shareholders' equity	(299)	1,145
Noncontrolling interests	255	240
Total equity	(44)	1,385
Total liabilities and equity	$ 11,958	$ 13,459

minority interest / non-controlling interests

In the consolidated profit and loss account minority interest is the proportion of the results for the year that relate to the minority holdings.

COLGATE-PALMOLIVE COMPANY

Consolidated Statements of Income

For the years ended December 31,

(Dollars in Millions Except Per Share Amounts)

	2015	2014	2013
Net sales	$ 16,034	$ 17,277	$ 17,420
Cost of sales	6,635	7,168	7,219
Gross profit	9,399	10,109	10,201
Selling, general and administrative expenses	5,464	5,982	6,223
Other (income) expense, net	62	570	422
Charge for Venezuela accounting change	1,084	—	—
Operating profit	2,789	3,557	3,556
Interest (income) expense, net	26	24	(9)
Income before income taxes	2,763	3,533	3,565
Provision for income taxes	1,215	1,194	1,155
Net income including noncontrolling interests	1,548	2,339	2,410
Less: Net income attributable to noncontrolling interests	164	159	169
Net income attributable to Colgate-Palmolive Company	$ 1,384	$ 2,180	$ 2,241
Earnings per common share, basic	$ 1.53	$ 2.38	$ 2.41
Earnings per common share, diluted	$ 1.52	$ 2.36	$ 2.38

Minority Interest

The formula for Enterprise Value is as follows:

Enterprise Value = Market value of common stock + Market value of preferred equity + Market value of debt + Minority interest – Cash

Impact of Minority Interest in Enterprise Value

1. The aim of adding minority interest to EV is to facilitate an "apples to apples" comparison between EV and figures such as Total Sales, EBIT, and EBITDA.
2. The equity value shown in the consolidated financial statement will always show the value of the parent company's stake in its subsidiaries. Thus, if the parent company owns 80% of its subsidiary and its subsidiary is worth $1,000, then the equity value reflects 80% of $1,000, or $800.
3. A problem arises because, according to accounting regulations, the parent company must show 100% of all other figures related to its subsidiaries if it owns more than 50% of its subsidiary.
4. Thus, the parent company doesn't reflect 80% of figures such as Total Sales, EBIT, and EBITDA. This creates a problem regarding the calculation of various valuation ratios such as EV/Total Sales, EV/EBIT, and EV/EBITDA because the numerator only reflects 80% of the subsidiary, whereas the denominator reflects 100% of the subsidiary.

5. It is usually not possible or practical to adjust the denominator because companies don't disclose enough information. Therefore, we adjust the numerator to reflect 100% of the subsidiary.

Associate

Associate refers to an entity in which the investor has significant influence but does not have full control like a parent and a subsidiary relationship. Usually, the investor has significant influence when it has 20% to 50% of shares of another entity.

Accounting for Investment in Associates

Accounting for investment in associates is done using the equity method. In the equity method, there is not a 100% consolidation used. Instead, the proportion of shares owned by the investor will be shown as an investment in accounting.

When an investor takes some shares in associate than in the balance sheet of the investor, it is recorded as an "increase in Associates," and cash gets reduced by the same amount. The dividend from the associate is shown as an increase in cash for the investor. To record the proportion of the net income of associate, the investment revenue of investor gets credit and investment in associate account get debited

Example of Investment in Associates

Let's say Corp ABC has purchased 30% shares of XYZ company. That means ABC has significant influence over XYZ, and XYZ can be treated as an associate of ABC. Value of 30% shares is $500,000. So, while making a purchase, below will be an accounting transaction for ABC.

Initial	Dr	Cr
Investment in Associates (Asset)	$500,000	
Cash		$500,000

After 6 months, XYZ declares $10,000 dividends to its shareholders. That means ABC will receive 30% of dividends or $3,000. Below will be accounting entries for the same:

After Dividends	Dr	Cr
Cash	$3,000	
Investment in Associates		$3,000

XYZ also declares a net income of $50,000. ABC will debit 30% of $50,000 in its "Investment in Associates" account while crediting the same amount as "Investment Revenue" in its Income Statement.

Accounting for Net Income	Dr	Cr
Investment in Associates	$15,000	
Investment Revenue		$15,000

Valuing Holdings in other firms

To value the holdings in other firms you would value the equity in each holding separately and estimate the value of the proportional holding. This would then be added on to the value of the equity of the parent company. Thus, to value a firm with minority holdings in three other firms, you would value the equity in each of these firms, take the percent share of the equity in each and add it to the value of equity in the parent company.

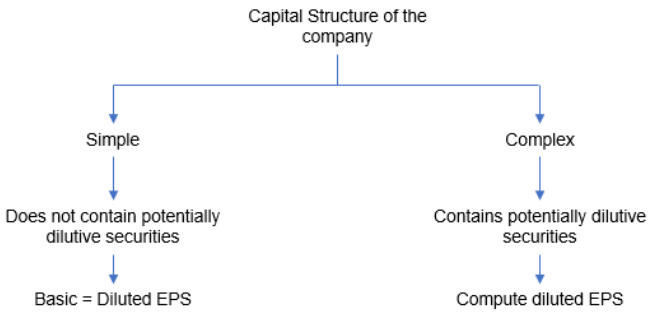
When income statements are consolidated, you would first need to strip the income, assets and debt of the subsidiary from the parent company's financials before you do any of the above. If you do not do so, you will double count the value of the subsidiary.

Why, you might ask, do we not value the consolidated firm? You could, and in some cases because of the absence of information, you might have to. The reason we would suggest separate valuations is that the parent and the subsidiaries may have very different characteristics – costs of capital, growth rates and reinvestment rates. Valuing the combined firm under these circumstances may yield misleading results. There is another reason. Once you have valued the consolidated firm, you will have to subtract out the portion of the equity in the subsidiary that the parent company does not own. If you have not valued the subsidiary separately, it is not clear how you would do this. Note that the conventional practice of netting out the minority interest does not accomplish this, because minority interest reflects book rather than market value.

As a firm's holdings become more numerous, estimating the values of the holdings will become more onerous. If the holdings are publicly traded, substituting in the market values of the holdings for estimated value is an alternative worth exploring. While you risk building into your valuation any mistakes the market might be making in valuing these holdings, this approach is more time efficient.

DILUTION OF EARNINGS PER SHARE (EPS)

- EPS is one of the most commonly used corporate **profitability performance measure**. It is reported only for shares of common stock also known as ordinary stock.



- **Dilutive securities** are financial instruments - Stock options, warrants convertible debt, convertible preferred stock which increases the number of common shares **if exercised**; this then reduces, or dilutes, the EPS.

- **Antidilutive securities** are stock financial instruments that would increase EPS if exercised or converted to common stock.

- Diluted EPS is a calculation used to gauge the quality of a company's earnings per share if all convertible securities were exercised. It will usually be lower than the simple or basic EPS but in the rare case that there are anti-dilutive securities it may be higher. In that case only the basic EPS is reported in the financial statements.

$$\text{Basic EPS} = \frac{\text{Net Income} - \text{Preferred dividends}}{\text{Weighted average number of common shares outstanding}}$$

$$\text{Diluted EPS} = \frac{\text{Adjusted income available to common shares}}{\text{Weighted average common and potential common shares outstanding}}$$

Example on Basic & Diluted EPS

A company has 1 lakh common shares and 10% debentures of Rs. 10 lakhs outstanding for the full year. The debentures are convertible to 1 lakh common shares. Compute the basic & diluted EPS if net profit earned after tax is Rs. 10 lakhs. It is given that tax rate is 30%.

Solution

$$\text{Basic EPS} = \frac{\text{Rs. 10 lakh}}{\text{1 lakh shares}} = \text{Rs 10 per share}$$

If debentures are converted to common shares, then interest cost (Net of tax) is saved

$$\text{Diluted EPS} = \frac{\text{Rs. 10 lakh} + \text{Rs. 1 lakh}(0.7)}{\text{1 lakh shares} + \text{1 lakh shares (Convertible debentures)}} = \text{Rs 5.35 per share}$$

DEFERRED TAX

- To understand deferred tax, it shall be noted at the outset that as per the accounting standards followed by companies, there are two different financial reports which an organization prepares every fiscal year – an income statement and a tax statement.

- The primary motive behind the creation of two different financial reports is varying guiding principles which govern the recording of companies' income and their taxation. In other words, the guidelines behind the preparation of a company's income statement and tax statement are slightly different. Hence, sometimes the numbers mentioned in both these reports might vary.

- It is this disparity that creates the scope for deferred tax. The term deferred tax, in essence, refers to the tax which shall either be paid or has already been settled due to transient inconsistency between an organization's income statement and tax statement.

- A deferred tax of any type (DTA/ DTL) is recorded in the balance sheet of an organization; however, their point of generation is typically in the income statement.

- When there is a difference in timing, between the amounts as per tax records (Based on Income Tax Act) and the amounts as per accounting records (Based on Accounting Standards), and such differences will reverse in future – then it is called a temporary difference and leads to creation of DTA/DTL.

- If the differences are not reversed, then it is called a permanent difference. It does not give rise to deferred tax assets or liabilities

Example: The depreciation method followed by companies and the Income Tax Department is different. Hence, a natural inconsistency appears between depreciation numbers in the income statement and tax statement. There may be different book and tax depreciation which may also arise due to difference in depreciation rates.

Suppose that Company A calculates depreciation in the straight-line method; whereas the Income Tax Department follows accelerated depreciation method. This company has machinery worth Rs. 1,00,000 with an estimated lifetime of 5 years and it expects no salvage value for the same. Hence, depreciation as per their calculation shall be Rs. 20,000. Now, as the Income Tax Department follows accelerated depreciation method, their calculation shows a depreciation of Rs. 25,000. As seen, depreciation considered by the IT department is higher, which implies a lower tax liability (Current tax) for the company for that particular fiscal year. However, company A will adjust depreciation in the subsequent years so that it eliminates such inconsistency between the two depreciation figures by the end of machinery's lifetime. Hence, it poses a future financial obligation accruing in that particular year – evident in the table below by deferred tax expense, thus creating a deferred tax liability.

This DTL will be reversed in the 5th year when the income tax department does not allow any depreciation and the deferred tax liability account will be settled.

Creation of DTL in the first year

Particulars	As per tax books	As per Accounts
Profit before depreciation	1,00,000	1,00,000
Depreciation (Temporary difference)	25,000	20,000
Profit after depreciation	75,000	80,000
Current tax (Tax rate: 40%)	30,000	30,000
Deferred tax expense	-	2,000
Overall tax impact	30,000	32,000
Profit after tax	45,000	48,000

VALUATION

Enterprise Value

Enterprise Value (EV) is the measure of a company's total value. It looks at the entire market value rather than just the equity value, so all ownership interests and asset claims from both debt and equity are included. EV can be thought of as the effective cost of buying a company or the theoretical price of a target company (before a takeover premium is considered).

The simple formula for enterprise value is:

EV = Market Capitalization + Market Value of Debt – Cash and Equivalents

The extended formula is:

EV = Common Shares + Preferred Shares + Market Value of Debt + Minority Interest – Cash and Equivalents

Why add minority interest to EV?

Non-controlling interest is the portion of a subsidiary not owned by the parent company (who owns a greater than 50% but less than 100% position in the subsidiary). The financial statements of this subsidiary are consolidated in the financial results of the parent company.

We add this minority interest to the calculation of EV because the parent company has consolidated financial statements with that minority interest; meaning the parent includes 100% of the revenues, expenses, and cash flow in its numbers even though it doesn't own 100% of the business.

By including the minority interest, the total value of the subsidiary is reflected in EV.

What Does Enterprise Value Tell You?

Enterprise value (EV) could be thought of like the theoretical takeover price if a company were to be bought. EV differs significantly from simple market capitalization in several ways, and many consider it to be a more accurate representation of a firm's value. The value of a firm's debt, for example, would need to be paid off by the buyer when taking over a company. As a result, enterprise value provides a much more accurate takeover valuation because it includes debt in its value calculation.

Why doesn't market capitalization properly represent a firm's value? It leaves a lot of important factors out, such as a company's debt on the one hand and its cash reserves on the other. Enterprise value is basically a modification of market cap, as it incorporates debt and cash for determining a company's valuation.

Market capitalization is not intended to represent a company's book value. Instead, it represents a company's value as determined by market participants.

VALUATION METHODS

Dividend Discount Model

The dividend discount model (DDM) is a quantitative method used for predicting the price of a company's stock based on the theory that its present-day price is worth the sum of all of its future dividend payments when discounted back to their present value. It attempts to calculate the fair value of a stock irrespective of the prevailing market conditions and takes into consideration the dividend payout factors and the market expected returns. If the value obtained from the DDM is higher than the current trading price of shares, then the stock is undervalued and qualifies for a buy, and vice versa.

Advantages - It is the most commonly used model to calculate share price and is therefore the easiest to understand. It values a company's stock without taking into account market conditions, so it is easier to make comparisons across companies of different sizes and in different industries.

Disadvantages - It does not take into account non dividend factors such as brand loyalty, customer retention and the ownership of intangible assets, all of which increase the value of a company. The Gordon Growth Model also relies heavily on the assumption that a company's dividend growth rate is stable and known

Dividends are appropriate as a measure of cash flow in the following cases:

- The company has a history of dividend payments.
- The dividend policy is clear and related to the earnings of the firm.
- The perspective is that of a minority shareholder

Firms in the mature stage of the industry life cycle are most likely to meet the first two criteria

Type of DDM

One-Period DDM

The one-period dividend discount model is a variation of the dividend discount model. The one-period dividend discount variation is used to determine the intrinsic value of a stock that is planned to be held for one period only (usually one year).

Similar to the general dividend discount model, the one-period variation is based on the assumption that the intrinsic value of a stock equals the sum of all future cash flows discounted back to its present value.

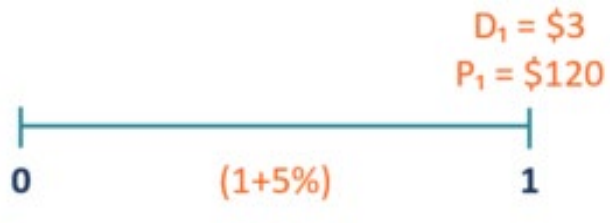
The mathematical formula that allows calculating the intrinsic value of a stock using the one-period dividend discount model is given below:

$$V_0 = \frac{D_1 + P_1}{1 + r}$$

Where:
V0 – the current fair value of a stock
D1 – the dividend payment in one period from now
P1 – the stock price in one period from now
r – the estimated cost of equity capital

Example

Assume that you are an investment analyst. Your client has asked you to assess the viability of their investment in ABC Corp. The client expects to hold the investment for one year only and sell it at the end of the holding period. You've discovered that ABC Corp. will pay $3 in dividends per share, while the selling price at the end of the holding period can reach $120 per share. The estimated cost of equity capital is 5%. Currently, the stock of ABC Corp. trades at $118 per share.



In order to assess the viability of the investment, you should determine the intrinsic value of the company's stock. It can be found using the one-period dividend discount model. By inputting the known variables into the formula above, the intrinsic stock value can be calculated in the following way:

$$V_0 = \frac{\$3 + \$120}{1 + 0.05} = \frac{\$123}{1.05} = \$117.14$$

The intrinsic value of the company's stock is $117.14, which is less than the company's current stock price ($118). Therefore, we can say that the stock is currently overvalued.

Multi Period DDM

A multiple-period dividend discount model is a variation of the dividend discount model. It is often used in situations when an investor is expecting to buy a stock and hold it for a finite number of periods and sell the stock at the end of the holding period.

Similar to the general dividend discount model, the multiple-period model is based on the assumption that the intrinsic value of a stock equals the sum of all future cash flows discounted back to their present values.

In such a scenario, an investor expects to hold the stock for the multiple periods. Thus, the future cash flows from the stock will include several dividend payments, as well as the expected selling price of the stock.

Subsequently, the current intrinsic value of a stock may be calculated by finding the sum of the future dividend payments and the anticipated selling price discounted back to their present values.

The mathematic formula that helps to calculate the fair value of a stock using the multiple-period dividend discount formula is given below:

$$V_0 = \frac{D_1}{1+r} + \frac{D_2}{(1+r)^2} + \cdots + \frac{D_n + P_n}{(1+r)^n}$$

where: V_0 = fundamental value
D_i = dividends expected to be received at end of year i, i = 1 to n

P_n = price expected upon sale at end of year n
r = required return on equity
n = length of holding period

The General Dividend Discount Model

If we extend the holding period indefinitely, the value simply becomes the present value of an infinite stream of dividends, represented by John Burr Williams's (1938) original DDM formula:

$$V_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t}$$

While the DDM is theoretically correct, applying it in practice requires the analyst to accurately forecast dividends for many periods, a task for which we rarely can expect to have sufficient information. We can use one of several growth models, including the:

- Gordon constant growth model
- Two-stage growth model
- Three-stage growth model

Gordon Growth Model

It assumes that dividends increase at a constant rate indefinitely

$$V_0 = D_0 * \frac{(1+g)^1}{(1+r)^1} + D_0 * \frac{(1+g)^2}{(1+r)^2} + \cdots + D_0 * \frac{(1+g)^n}{(1+r)^n}$$

$$V_0 = D_0 * \frac{1+g}{r-g} = \frac{D_1}{r-g}$$

Where,
V_0 =fundamental value
D_0 = dividend just paid
D_1 = dividends expected to be received at end of Year 1
r = required return on equity
g = dividend growth rate

The model assumes that:

- The firm expects to pay a dividend, D1, in one year.
- Dividends grow indefinitely at a constant rate, g (which may be less than zero)
- The growth rate, g, is less than the required return, r

For most companies, the Gordon growth model assumption of constant dividend growth that continues into perpetuity is unrealistic. For example, many companies experience growth rates in excess of the required rate of return for short periods of time as a result of a competitive advantage they have developed. We need more realistic multistage growth models to estimate value for companies with several stages of future growth. However, whichever multistage model we use, there are two important points to keep in mind:

- We're still just forecasting dividends into the future and discounting them back to today to find intrinsic value.
- Over the long term, growth rates tend to revert to a long-run rate approximately equal to the long-term growth rate in real gross domestic product (GDP) plus the long-term inflation rate.

3 Stage DDM

Three-stage models are appropriate for firms that are expected to have three distinct stages of earnings growth. For example, suppose we forecast that a biotech company will experience supernormal growth of 25% for three years, then 15% growth for five years, and finally slow down to a stable, long-run rate of 3%

While the basic GGM assumes constant growth, most firms go through a pattern of growth that includes several phases:

- An initial growth phase, where the firm has rapidly increasing earnings, little or no dividends, and heavy reinvestment.
- A transition phase, in which earnings and dividends are still increasing but at a slower rate as competitive forces reduce profit opportunities and the need for reinvestment.
- A mature phase, in which earnings grow at a stable but slower rate, and payout ratios are stabilizing as reinvestment matches depreciation and asset maintenance requirements.

DCF Valuation

FCFF and FCFE

- Cash flows into the firm in the form of revenue as it sells its product, and cash flows out as it pays its cash operating expenses (e.g., salaries and taxes, but not interest expense, which is a financing and not an operating expense).
- The firm takes the cash that's left over and makes short-term net investments in working capital (e.g., inventory and receivables) and long-term investments in property, plant, and equipment (PP&E).
- The cash that remains is available to pay out to the firm's investors: bondholders and common shareholders and is called free cash flow to the firm (FCFF) because it's free to pay out to the firm's

FCFF is calculated as:

FCFF = EBIT(1-t) + Depreciation - Capex – Increase in Working Capital

FCFE is the cash available to common shareholders after funding capital requirements, working capital needs, and debt financing requirements. It is calculated as -

FCFE = FCFF- Int(l -tax rate) + net borrowing

FCFF vs FCFE

- The differences between FCFF and FCFE account for differences in capital structure and consequently reflect the perspectives of different capital suppliers.
- FCFE is easier and more straightforward to use in cases where the company's capital structure is not particularly volatile.
- On the other hand, if a company has negative FCFE and significant debt outstanding, FCFF is generally the best choice.

- We can always estimate equity value indirectly by discounting FCFF to find firm value and then subtracting out the market value of debt to arrive at equity value.

Free Cash Flows vs Dividends

Analysts often prefer to use free cash flow rather than dividend-based valuation for the following reasons:

- Many firms pay no, or low, cash dividends
- Dividends are paid at the discretion of the board of directors. It may, consequently, be poorly aligned with the firm's long-run profitability
- Free cash flows may be more related to long-run profitability of the firm as compared to dividends

Valuation

The value of the firm is the present value of the expected future FCFF discounted at the WACC
Enterprise Value/ Firm value = FCFF discounted at the WACC

The value of the firm's equity is the present value of the expected future FCFE discounted at the required return on equity:

Equity value = FCFE discounted at the required return on equity

Given the value of the firm, we can also calculate equity value by simply subtracting out the net debt:
Equity value = firm value- net debt

Calculation of Discount Rates

Cost of Equity

A firm's cost of equity represents the compensation the market demands in exchange for owning the asset and bearing the risk of ownership. It is usually calculated using the CAPM (Capital Asset Pricing Model)

By CAPM

No matter how much you diversify your investments, some level of risk will always exist. So, investors naturally seek a rate of return that compensates for that risk. The capital asset pricing model (CAPM) helps to calculate investment risk and what return on investment an investor should expect

The model starts with the idea that individual investment contains two types of risk:

- ***Systematic Risk*** – These are market risks—that is, general perils of investing—that cannot be diversified away. Interest rates, recessions, and wars are examples of systematic risks.
- ***Unsystematic Risk*** – Also known as "specific risk," this risk relates to individual stocks. In more technical terms, it represents the component of a stock's return that is not correlated with general market moves.

Modern portfolio theory shows that specific risk can be removed or at least mitigated through diversification of a portfolio. The trouble is that diversification still does not solve the problem of systematic risk; even a portfolio holding all the shares in the stock market can't eliminate that risk. Therefore, when calculating a deserved return, systematic risk is what most plagues investors

CAPM evolved as a way to measure this systematic risk. Sharpe found that the return on an individual stock, or a portfolio of stocks, should equal its cost of capital. The standard formula remains the CAPM, which describes the relationship between risk and expected return.

$$Ke = Rf + \beta*(Rm-Rf)$$

Where,
Ke = cost of equity
Rf = Risk free rate
Rm = Market Return
Rm-Rf = Equity Risk Premium
β = Beta, measure of systematic risk

Beta Calculation

Beta is a measure of a stock's volatility in relation to the overall market. By definition, the market, such as the S&P 500 Index, has a beta of 1.0, and individual stocks are ranked according to how much they deviate from the market.

A stock that swings more than the market over time has a beta above 1.0. If a stock moves less than the market, the stock's beta is less than 1.0

For e.g. – A beta of 2 implies that if the market moves by 1%, the stock will move by 2% in the same direction.

$$\textit{Formula for Beta of stock x} = \frac{Cov(Rm,Rx)}{Var(Rm)}$$

How to calculate Beta for Non-Public companies –
Identify Benchmark company and find its beta and then unlever the beta –

$$\text{Unlevered beta} = beta * \left(\frac{1}{1+(1-t)*\frac{D}{E}}\right)$$

Lever up the beta using debt and equity estimates of your company

$$\text{Levered Beta} = Unlevered\ Beta * \left(1 + (1-t) * \left(\frac{D}{E}\right)\right)$$

How to calculate Risk Premium

The equity risk premium is the return in excess of the risk-free rate that investors require for holding equity securities. It is usually defined as the difference between the required return on a broad equity market index and the risk-free rate:

I. **Historical estimates** – It is the difference between the historical mean return for a broad-based equity-market index and a risk-free rate over a given time period
II. **Forward looking estimates** –
Gordon Growth Model
GGM equity risk premium = (1-year forecasted dividend yield on market index) + (consensus long-term earnings growth rate) - (long-term government bond yield)
III. **Survey/analyst's estimates** - use the consensus of the opinions from a sample of people use the consensus of the opinions from a sample of people.

Build up Method - The method simply adds a risk premium for the added risk arising from holding the firm's equity to the yield to maturity (YTM) of the company's long-term debt

Cost of Debt

There are a couple of different ways to calculate a company's cost of debt, depending on the information available.

The formula (risk-free rate of return + credit spread) multiplied by (1 - tax rate) is one way to calculate the after-tax cost of debt.

For example, say the risk-free rate of return is 1.5% and the company's credit spread is 3%. Its pre-tax cost of debt is 4.5%. If its tax rate is 30%, then the after-tax cost of debt is 3.15% = [(0.015 + 0.03) × (1 - 0.3)].

As an alternative way to calculate the after-tax cost of debt, a company could determine the total amount of interest that it is paying on each of its debts for the year. The interest rate that a company pays on its debts is inclusive of both the risk-free rate of return and the credit spread from the formula above because the lender(s) will take both into account when initially determining an interest rate.

Once the company has its total interest paid for the year, it divides this number by the total of all of its debt. This is the company's average interest rate on all of its debt. The after-tax cost of debt formula is the average interest rate multiplied by (1 - tax rate).

For example, say a company has a $1 million loan with a 5% interest rate and a $200,000 loan with a 6% rate. The average interest rate, and its pre-tax cost of debt, is 5.17% = [($1 million × 0.05) + ($200,000 × 0.06)] ÷ $1,200,000. The company's tax rate is 30%. Thus, its after-tax cost of debt is 3.62% = [0.0517 × (1 - 0.30)].

WACC

WACC - overall required rate of return for those who supply a company with capital.

$$\textbf{WACC} = \frac{Market\ Value\ of\ Debt}{Market\ Value\ of\ Debt\ and\ Equity} * rd * (1 - t) + \frac{Market\ Value\ of\ Equity}{Market\ Value\ of\ Debt\ and\ Equity} * re$$

The discount rate should correspond to the type of cash flow being discounted. Cash flows to the entire firm should be discounted with the WACC. Alternatively, cash flows in excess of what is required for debt service should be treated as cash flows to equity and discounted at the required return to equity.

Forecasting of Cashflows

Like DDM, the cashflows are forecasted using a multi stage model.

The simple free cash flow model, in which we forecast total FCFE or FCFF, looks a lot like the multistage dividend discount models. The benefit of using free cash flow models, however, is when we refine our

approach by forecasting the values and/or growth rates in the components of free cash flow over the first stage and then calculate free cash flow in each year using one of our formulas.

We would use a two-stage model for a firm with two stages of growth: a short-term supernormal growth phase and a long-term stable growth phase. For example, a firm with a valuable patent that expires in seven years might experience a high growth rate for seven years and then immediately drop to a long-term, lower growth rate beginning in the eighth year. We would use a three-stage model for a firm that we expect to have three distinct stages of growth (e.g., a growth phase, a mature phase, and a transition phase).

Terminal Value

There are two basic approaches for calculating terminal value: using a single-stage model or a multiple approach:

Using Single Stage Model

$$\textit{Terminal Value in year n} = \frac{FCFF_{(n+1)}}{WACC-g}$$

Where g is the rate at which we expect the stock to grow perpetually.

Multiple Approach

The other way to do this is to use valuation multiples (like P/E ratios) to estimate terminal value. The terminal value in year n in terms of P/E, for example, would be expressed as:

terminal value in year n = (trailing P/E) x (earnings in year n)

terminal value in year n= (leading P/E) x (forecasted earnings in year n+1)

Steps to DCF

1. Forecasting Cashflows – The cashflows are forecasted by forecasting the components that make FCFF. They are forecasted usually for 5-10 years depending on the company and the growth stage it is at
2. Calculate the discount rate – If FCFE is used, calculate ke of the firm using the CAPM model or any such suitable method. If FCFF is used, the cashflows need to be discounted using WACC (Weighted avg cost of capital)
3. Calculate the terminal value – If cashflows are forecasted for 5 years, the terminal value can be calculated as on year 5 and the be discounted to year 0 along with other cashflows. The terminal value can be calculated by dividing the last cashflow forecast by the difference between discount rate and terminal growth rate or by using the multiple approach as discussed above
4. By adding the sum of the discounted FCFF and terminal value, we arrive at the enterprise value. We can subtract net debt to arrive at the equity value. In case FCFE has been forecasted, the sum of the discounted FCFEs will be equal to the equity value.

Advantages of DCF

- DCF Valuation truly captures the underlying fundamental drivers of a business (cost of equity, weighted average cost of capital, growth rate, re-investment rate, etc.). Consequently, this comes closest to estimating intrinsic value of the asset/business.

- Unlike other valuations, DCF relies on Free Cash Flows. To a larger extent, Free Cash Flows (FCF) are a reliable measure that eliminate the subjective accounting policies and window dressing involved in reported earnings
- Besides explicitly considering the business drivers involved, DCF allows investors to incorporate key changes in the business strategy in the valuation model, which will otherwise not be considered in other valuation models (like relative, APV, etc.)
- While other methods like relative valuation are fairly easier to calculate, their reliability becomes questionable when the entire sector or market is over-valued or under-valued. DCF cuts across through this quandary and predicts the best possible intrinsic value.

Disadvantage of DCF

- DCF Valuation is extremely sensitive to assumptions related to perpetual growth rate and discount rate. Any minor tweaking here and there, and the DCF Valuation will fluctuate wildly and the fair value so generated won't be accurate.
- It works best only when there is a high degree of confidence about future cash flows. But if the company's operations lack visibility, it becomes difficult to predict sales, operating expenses and capital investment with certainty
- One major criticism of DCF is that the terminal value comprises far too much of the total value (65-75%). Even a minor variation in the assumptions on terminal year can have a significant impact on the final valuation

Relative Valuation

A relative valuation model is a business valuation method that compares a company's value to that of its competitors or industry peers to assess the firm's financial worth. Relative valuation models are an alternative to absolute value models, which try to determine a company's intrinsic worth without any reference to another company or industry average. The investors may use relative valuation models when determining whether a company's stock is a good buy.

Advantages of Relative valuation

- The allure of using multiples is the ease of use, and it is a much quicker way to value a company. The calculation of the ratios usually consists simply of a simple arithmetic operation, usually a division, without going into the complexity of the calculation of the cash flow discount.
- The intuitive nature of relative valuation is attractive to prospective investors than the technical nature of DCF.

Disadvantages of Relative valuation

- It is difficult to find true value for some companies, because of low trading pattern and small market capitalization.
- Another issue with using multiples for valuation is it builds in errors that the market might be making in valuing any of the comparable companies. For example, if the market has overvalued all computer software companies, then using the average or median PE ratio of these companies will lead to us overvaluing the company.
- There is no assurance that the "cheaper" company will outperform its peer.

Common Multiples

- There are two main types of valuation multiples
  - **Equity Multiples** – These use prices of the company's share. For example, P/E, P/B, etc.
  - **Enterprise Value Multiples** – These use EV of the company. For example, EV/EBIT, EV/ EBITDA, etc.

  Equity level ratios can be affected by changes in the capital structure without any change in the Enterprise value. Enterprise level ratios help compare companies with different leverage ratios. Also, Enterprise level ratios are generally less affected by accounting changes/ differences as the denominator is higher than that in the case of equity ratios.

- Some of the common metrics used in this type of valuation include
  - **Price to Earnings or P/E** – The price-to-earnings ratio (P/E ratio) is the ratio for valuing a company that measures its current share price relative to its per-share earnings (EPS). P/E may be estimated on a trailing (backward-looking) or forward (projected) basis and is the most widely used tool by which investors and analysts determine a stock's relative valuation. It is read with expected growth to assess how stocks are relatively valued. The lower the P/E ratio, the more attractive the investment.

  - **Price to Book or P/B** – Companies use the price-to-book ratio (P/B ratio) to compare a firm's market capitalization to its book value. It's calculated by dividing the company's stock price per share by its book value per share (BVPS). It is often read with ROE. A stock with lower P/B ratio and higher ROE is valued higher.

  - **Price to Sales or P/S** – The price-to-sales ratio (Price/Sales or P/S) is calculated by taking a company's market capitalization (the number of outstanding shares multiplied by the share price) and divide it by the company's total sales or revenue over the past 12 months. It is assessed with net margin (Higher is better). The lower the P/S ratio, the more attractive the investment.

  - **Enterprise Value to EBITDA or EV/EBITDA** – Investors and analysts use the enterprise value (EV) metric to calculate a company's total monetary value or assessed worthwhile EBITDA is a useful way to measure a company's overall financial performance and profitability. The ratio is used to compare the value of a company, debt included, to the company's cash earnings less non-cash expenses. It's ideal for analysts and investors looking to compare companies within the same industry. This ratio is read along with reinvestment rate to judge how stocks are relatively valued. A stock with lower EV/EBITDA and low reinvestment needs is attractive.

  - **EV/Sales** – Enterprise value-to-sales is a financial valuation measure that compares the enterprise value of a company to its annual sales. Generally, a lower EV/sales multiple will indicate that a company may be more attractive or undervalued in the market.

- Understanding relative valuation using PE multiple: A company with a high P/E ratio is trading at a higher price per dollar of earnings than its peers and is considered overvalued. Likewise, a company with a low P/E ratio is trading at a lower price per dollar of EPS and is considered undervalued. This framework can be carried out with any multiple of price to gauge relative market value.

  Therefore, if the average P/E for an industry is 10 times and a particular company in that industry is trading at 5 times of earnings, it is relatively undervalued to its peers.

- Estimating Relative Value of Stock: In addition to providing a gauge for relative value, the P/E ratio allows analysts to back into the price that a stock should be trading at based on its peers.

  Example: If the average P/E for an industry is 20, it means the average price of stock from a company in the industry trades at 20 times its EPS.

  Company A trades for Rs. 50 in the market and has an EPS of Rs. 2. The P/E ratio is clearly 25. This is higher than the industry average of 20, which means Company A is overvalued. If Company A were trading at 20 times its EPS, the industry average, it would be trading at a price of Rs. 40, which is the relative value. In other words, based on the industry average, Company A is trading at a price that is Rs. 10 higher than it should be, representing an opportunity to sell.

Comparison of P/E ratio and EV/EBITDA

P/E ratio is widely known and used as an indicator of a company's future growth potential. However, it does not reveal a full picture, and it is most useful when comparing only companies within the same industry or comparing companies against the general market.

There are problems that arise for investors with the use of the P/E ratio. The stock price can get run up if investors are overly optimistic causing an overvalued P/E ratio. Also, the earnings portion of the metric can be manipulated somewhat.

The EV/EBITDA ratio helps to allay some of the P/E ratios downfalls and is a financial metric that measures the return a company makes on its capital investments. EBITDA provides a clearer picture of the financial performance of a company since it strips out debt costs, taxes, and accounting measures like depreciation, which spreads the costs of fixed assets out for many years. The other component, EV is the sum of a company's equity value or market capitalization plus its debt less cash. It is typically used in buyouts. The EV/EBITDA is calculated to achieve an earnings multiple that is more comprehensive than the P/E ratio.

However, the EV/EBITDA ratio has its drawbacks, such as the fact that it doesn't include capital expenditures, which for some industries can be significant. As a result, it may produce a more favourable multiple by not including those expenditures.

The P/E ratio has been established as a prime market valuation metric, and the sheer volume of current and historical data gives the metric weight in regard to stock analysis. Some analysts contend that using the EV/EBITDA ratio versus the P/E ratio as a valuation method produces better investment returns.

Both metrics have inherent advantages and disadvantages. As with any financial metric, it's important to consider several financial ratios including the P/E ratio and the EV/EBITDA ratio in determining whether a company is fairly valued, overvalued, or undervalued.

Industry Specific Ratios

- Cement – EV/Ton
- Retail – Same store sales growth (SSSG), New Store Growth (Revenue); Per Sq. Ft. Sales (Store productivity) say compare DMART > BigBazaar; EV/EBITDA, EV/EBIDTAR
- Banking - P/BV, P/E
- Insurance – P/B, P/NAV, P/Embedded Value

- Power - EV/EBITDA, EV/BV
- FMCG- EV/EBITDA, EV/Sales
- Automobile- EV/EBITDA
- Telecom -ARPU
- Healthcare Hospitals – Patient bed Days (Occupancy rate, no of beds, no. of patients); Contract/Own – Lease (different D&A); EV/Operating bed, EV/EBIDTAR, ALOS
- Pharma – EV/EBITDA, PEG (Size different, product mix similar); MRs productivity (Operations); Labs – Same lab revenue growth + new growth
- Cab Aggregators – Passenger Ride Km, EV/No. of rides completed, EV/Gross ride value
- Restaurant Aggregators – EV/Gross sales Value
- Online /Ecommerce – GMV
- Real Estate – EV/NAV
- Social Media – EV/Daily Monthly Users, EV/Next 12 month sales, EV/last 12 month sales

Multiple	Sector	Comments
EV/ Revenue	Various	Early Stage Companies
EV/ Subscriber	Various	Subscriber Based Business such as cable and DTH
EV/ EBITDA	Various	Many industrial and consumer industries, but not bank, insurance, Oil, Gas, Retail
EV/EBITA	Various	Commonly used in several media industry sub-sectors, gaming, chemicals, bus and retail Used when EBITDA multiples are less relevant due to significant differences in asset financing (e.g. – mix of leases, rentals, ownership)
EV/EBITDAX	Oil & Gas	Excludes exploration expenses
EV/ EBITDAR	Retail, Airlines	Used when there are significant rental and lease expenses incurred by business operations
P/BV	Tech/Banks/Insurance	Banks shareholder equity is important because it is looked as a buffer/protection for depositors
EV/FFO	Real Estate	Principally used in US
P/E	Various	Often using normalized cash earnings, excluding both exceptional items & goodwill amortization
PEG Ratio	High tech, high growth	Big difference in growth across companies

Private Company Valuation

The process of valuing private companies is not different from the process of valuing public companies. You estimate cash flows, attach a discount rate based upon the riskiness of the cash flows and compute a present value.
As with public companies, you can either value

- The entire business, by discounting cash flows to the firm at the cost of capital.
- The equity in the business, by discounting cashflows to equity at the cost of equity.

When valuing private companies, you face two standard problems:

- There is not market value for either debt or equity
- The financial statements for private firms are likely to go back fewer years, have less detail and have more holes in them.

Valuation Methods

Market Approach

Because it is difficult to establish private company valuation multiples, the most common approach is to use comparable company analysis (CCA). This approach, the appraiser searches for publicly-traded companies that closely resemble the subject company.

Using this approach, public companies in the same industry of a similar size, age, and growth rate are identified, and averages of their multiples or valuations are calculated for comparison to the subject company. This gives the appraiser an idea of where the subject private company fits within the industry and how it compares to its competitors.

Income Approach

The basis of the income or present value approach is the premise that the subject company's current full cash value is equal to the present value of future cash flows it will provide over its remaining economic life.

Estimating the revenue growth of the subject company by averaging the revenue growth rates of the comparable companies and then adjusting for company specific factors is the first step. After estimating revenue growth, expected changes in operating expenses, taxes, and working capital are estimated. Once all of these estimates are completed, free cash flow is calculated, providing the operating cash remaining after the deduction of expenditures. The next step is calculating the average beta (measure of market risk, disregarding debt), tax rates, and debt-to-equity (D/E) ratios of comparable companies, and ultimately, the weighted average cost of capital (WACC).

WACC - For a private company valuation, the cost of debt can be determined by examining the subject company's credit history and the interest rates being charged to the company. Equity can be estimated using the capital asset pricing model (CAPM). The debt and equity ratings and the cost of capital for comparable companies are also factored into WACC calculations. Beta section can be referred to see how it can be calculated for private firms.

Asset Based Approach

The asset-based approach estimates the value of firm equity as the fair value of its assets minus the fair value of its liabilities. It is generally not used for going concerns. Because it is easier to find comparable data at the firm level compared to the asset level, the income and market approaches would be preferred when valuing going concerns. Additionally, it is difficult to find data for individual intangible assets and specialized assets. Of the three approaches, the asset-based approach generally results in the lowest valuation because the use of a firm's assets in combination usually results in greater value creation than each of its parts individually.

The asset-based approach might be appropriate in the following circumstances:

- Firms with minimal profits and little hope for better prospects. In this situation, the firm might be valued more highly for its liquidation value rather than as a going concern by a firm that can put the assets to better use.
- Finance firms such as banks, where their asset and liability values (loan and security values) can be based on market prices and factors.
- Small companies or early-stage companies with few intangible assets.
- Natural resource firms where assets can be valued using comparable sales

Applying Discounts

Regardless of the method used for estimating the subject private company's valuation, several discounts need to be taken into consideration and applied where appropriate:

- **Marketability discount:** This discount considers the lack of ability to rapidly convert an ownership stake to cash.
- **Key man discount:** This discount might not always apply; it depends on the nature of the business. If the business is in a stable industry and already well-developed, losing a key employee or leader might not have any impact. But if the company is young or there is a lot of value associated with a particular individual, the impact of that person leaving could be substantial. Apple provides the best example: After Steve Jobs was forced out of the company, Apple experienced several disastrous years, only recovering after Jobs was brought back onboard. The value of Apple without Steve Jobs was substantially less than it was when he was leading the company.
- **Control discount:** If a minority stake in a private company is sold, a value adjustment needs to be made to account for this lack of operational and financial control. This can also apply in a public company but will have a much smaller impact; the impact in private companies is much more significant due to the lower level of transparency associated with private businesses.